

## Volume Iii Functions Continuity And Differentiation Reference

# Contents

|        |                                                 |    |
|--------|-------------------------------------------------|----|
| 0.1    | Hyperbolic Functions . . . . .                  | 3  |
| 0.2    | The Logarithm . . . . .                         | 4  |
| 0.3    | Power Functions and the Exponential . . . . .   | 4  |
| 0.4    | Trigonometric Functions . . . . .               | 5  |
| 0.5    | Limits . . . . .                                | 7  |
| 0.6    | Point Continuity . . . . .                      | 10 |
| 0.7    | Global Theorems . . . . .                       | 11 |
| 0.8    | Uniform Continuity . . . . .                    | 12 |
| 0.9    | Monotone Functions . . . . .                    | 13 |
| 0.10   | Limits at Infinity . . . . .                    | 14 |
| 0.11   | Approximation . . . . .                         | 14 |
| 0.12   | Gauge . . . . .                                 | 15 |
| 0.13   | The Tangent Problem . . . . .                   | 16 |
| 0.14   | The Derivative . . . . .                        | 16 |
| 0.14.1 | First Consequences . . . . .                    | 17 |
| 0.15   | One-Sided Derivatives . . . . .                 | 17 |
| 0.16   | Carathéodory and the Chain Rule . . . . .       | 18 |
| 0.16.1 | Carathéodory Characterization . . . . .         | 18 |
| 0.17   | The Shape Questions . . . . .                   | 18 |
| 0.17.1 | Interior Extrema . . . . .                      | 19 |
| 0.18   | The Mean Value Theorems . . . . .               | 19 |
| 0.19   | Reading the Graph from $f'$ and $f''$ . . . . . | 20 |
| 0.19.1 | First-Order Shape . . . . .                     | 20 |
| 0.19.2 | Second-Order Shape . . . . .                    | 21 |
| 0.19.3 | Higher-Order Derivatives . . . . .              | 21 |
| 0.19.4 | Second-Order Shape . . . . .                    | 22 |
| 0.19.5 | Darboux's Theorem . . . . .                     | 22 |
| 0.19.6 | Smoothness Classes . . . . .                    | 22 |
| 0.20   | The Algebra of Derivatives . . . . .            | 23 |
| 0.20.1 | From Characterization to Computation . . . . .  | 23 |
| 0.20.2 | Closure . . . . .                               | 24 |
| 0.20.3 | Global Forms on an Interval . . . . .           | 24 |
| 0.20.4 | Inverting the Operation . . . . .               | 25 |
| 0.20.5 | Indeterminate Forms: L'Hôpital's Rule . . . . . | 26 |
| 0.21   | Taylor: Construction and Its Error . . . . .    | 26 |
| 0.21.1 | The Differential . . . . .                      | 27 |

# Elementary Functions

## 0.1 Hyperbolic Functions

### Hyperbolic Functions

**Definition 0.1.1** (Hyperbolic Functions). For all  $x \in \mathbb{R}$ :

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional:  $\operatorname{sech} x := 1/\cosh x$ ,  $\operatorname{csch} x := 1/\sinh x$ ,  $\operatorname{coth} x := \cosh x/\sinh x$ .

**Proposition 0.1.2** (Hyperbolic Pythagorean Identity). For all  $x \in \mathbb{R}$ :

$$\cosh^2 x - \sinh^2 x = 1.$$

*Go to proof.*

**Proposition 0.1.3** (Addition Formulas for Hyperbolic Functions). For all  $x, y \in \mathbb{R}$ :

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y, \quad \cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

*Go to proof.*

**Proposition 0.1.4** (Limits of  $\tanh$ ).

$$\lim_{x \rightarrow +\infty} \tanh x = 1, \quad \lim_{x \rightarrow -\infty} \tanh x = -1.$$

*Go to proof.*

**Definition 0.1.5** (Inverse Hyperbolic Functions). Since  $\sinh$  is strictly increasing and bijective  $\mathbb{R} \rightarrow \mathbb{R}$ , its inverse  $\operatorname{arcsinh}$  is defined on all of  $\mathbb{R}$ . Since  $\cosh$  is strictly increasing on  $[0, \infty)$  with range  $[1, \infty)$ , its inverse  $\operatorname{arccosh}$  is defined on  $[1, \infty)$ . Since  $\tanh$  is strictly increasing with range  $(-1, 1)$ ,  $\operatorname{arctanh}$  is defined on  $(-1, 1)$ . Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad x \in \mathbb{R}.$$

$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1.$$

$$\operatorname{arctanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

## 0.2 The Logarithm

### The Logarithm

**Definition 0.2.1** (Natural Logarithm). The **natural logarithm**  $\ln : (0, \infty) \rightarrow \mathbb{R}$  is the inverse of the exponential function: for  $x > 0$ ,

$$\ln x := \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently,  $\ln x$  is defined by the integral

$$\ln x := \int_1^x \frac{1}{t} dt.$$

**Proposition 0.2.2** (Fundamental Logarithmic Limit).

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1.$$

*Go to proof.*

**Proposition 0.2.3** (Every Power Dominates the Logarithm). *For every  $r > 0$ :*

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^r} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} x^r \ln x = 0.$$

*Go to proof.*

## 0.3 Power Functions and the Exponential

### Power Functions and the Exponential

**Definition 0.3.1** (Power Function). Let  $n \in \mathbb{N}$ . The **integer power function** is  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) := x^n$ . For  $r \in \mathbb{R}$  and  $x > 0$ , the **real power** is defined by

$$x^r := e^{r \ln x}.$$

**Definition 0.3.2** (The Number  $e$ ). The **Euler number** is

$$e := \sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828 \dots$$

Both expressions define the same real number.

**Definition 0.3.3** (Exponential Function). The **exponential function**  $\exp : \mathbb{R} \rightarrow (0, \infty)$  is defined by the absolutely convergent power series

$$e^x := \exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}.$$

**Proposition 0.3.4** (Fundamental Exponential Limit).

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$$

*Go to proof.*

**Proposition 0.3.5** (Compound Interest Limit).

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

*Go to proof.*

**Proposition 0.3.6** (Exponential Dominates Every Power). *For every  $r > 0$ :*

$$\lim_{x \rightarrow \infty} \frac{x^r}{e^x} = 0.$$

*Go to proof.*

## 0.4 Trigonometric Functions

### Trigonometric Functions

**Definition 0.4.1** (Sine and Cosine via Power Series). The **sine** and **cosine** functions are defined for all  $x \in \mathbb{R}$  by the absolutely convergent power series

$$\begin{aligned}\sin x &:= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \\ \cos x &:= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\end{aligned}$$

**Proposition 0.4.2** (Pythagorean Identity). *For all  $x \in \mathbb{R}$ :*

$$\sin^2 x + \cos^2 x = 1.$$

*Go to proof.*

**Proposition 0.4.3** (Addition Formulas). *For all  $x, y \in \mathbb{R}$ :*

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \quad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

*Go to proof.*

**Definition 0.4.4** (Tangent, Cotangent, Secant, Cosecant).

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$  and  $\sec x$  are defined where  $\cos x \neq 0$ , i.e.,  $x \neq \pi/2 + k\pi$  for  $k \in \mathbb{Z}$ .

**Definition 0.4.5** (Inverse Trigonometric Functions). •  $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$  is the inverse of  $\sin|_{[-\pi/2, \pi/2]}$ .

- $\arccos : [-1, 1] \rightarrow [0, \pi]$  is the inverse of  $\cos|_{[0, \pi]}$ .
- $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$  is the inverse of  $\tan|_{(-\pi/2, \pi/2)}$ .

**Theorem 0.4.6** (Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

*Go to proof.*

**Proposition 0.4.7** (Second Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

*Go to proof.*

## Proofs

# Continuity

## 0.5 Limits

### Limits of Functions

**Definition 0.5.1** ( $\varepsilon$ -Closeness of a Function to a Value). Let  $X \subseteq \mathbb{R}$ , let  $f : X \rightarrow \mathbb{R}$ , let  $L \in \mathbb{R}$ , and let  $\varepsilon > 0$ . The function  $f$  is  **$\varepsilon$ -close to  $L$**  if  $|f(x) - L| \leq \varepsilon$  for all  $x \in X$ .

**Definition 0.5.2** (Local  $\delta$ -Closeness Near a Point). Let  $f : X \rightarrow \mathbb{R}$ , let  $L \in \mathbb{R}$ , let  $x_0$  be adherent to  $X$ , and let  $\varepsilon, \delta > 0$ . The function  $f$  is  **$\varepsilon$ -close to  $L$  near  $x_0$  within  $\delta$**  if

$$\forall x \in X \ (|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon).$$

**Definition 0.5.3** (Limit of a Function). Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . A real number  $L$  is the **limit of  $f$  at  $c$** , written  $\lim_{x \rightarrow c} f(x) = L$ , if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Go to uniqueness proof.*

**Definition 0.5.4** (Sequential Definition of Limit). Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The limit  $\lim_{x \rightarrow c} f(x) = L$  in the **sequential sense** holds if for every sequence  $(x_n) \subseteq A \setminus \{c\}$  with  $x_n \rightarrow c$ , the sequence  $(f(x_n))$  converges to  $L$ .

**Definition 0.5.5** (Neighbourhood Form of Limit). Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The limit  $\lim_{x \rightarrow c} f(x) = L$  in the **neighbourhood sense** holds if for every  $\varepsilon$ -neighbourhood  $V_\varepsilon(L)$  there exists a deleted  $\delta$ -neighbourhood  $V_\delta^*(c)$  such that

$$x \in V_\delta^*(c) \cap A \implies f(x) \in V_\varepsilon(L).$$

**Proposition 0.5.6** (Neighbourhood Form of the Limit). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  if and only if for every  $\varepsilon$ -neighbourhood  $V_\varepsilon(L)$  there exists a  $\delta$ -neighbourhood  $V_\delta(c)$  such that  $x \in V_\delta^*(c) \cap A$  implies  $f(x) \in V_\varepsilon(L)$ .*

*Go to proof.*

**Theorem 0.5.7** (Uniqueness of Limits of Functions). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $f$  has at most one limit at  $c$ .*

*Go to proof.*

**Proposition 0.5.8** (Sequential Criterion for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \setminus \{c\}$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Go to proof.*

**Corollary 0.5.9** (Three Equivalent Forms of the Function Limit). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The  $\varepsilon$ - $\delta$  form of  $\lim_{x \rightarrow c} f(x) = L$ , the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

*Go to proof.*

**Theorem 0.5.10** (Limit Implies Local Boundedness). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  for some  $L \in \mathbb{R}$ , then there exist  $\delta, M > 0$  such that  $|f(x)| \leq M$  for all  $x \in A$  with  $0 < |x - c| < \delta$ .*

*Go to proof.*

**Theorem 0.5.11** (Bounded Limits). *Let  $f : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ , and let  $a, b \in \mathbb{R}$ . If  $a \leq f(x) \leq b$  for all  $x \in A$  with  $x \neq c$ , and  $\lim_{x \rightarrow c} f(x) = L$ , then  $a \leq L \leq b$ .*

*Go to proof.*

**Theorem 0.5.12** (Squeeze Theorem for Function Limits). *Let  $f, g, h : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Suppose  $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} g(x) = L$ , and suppose there exists  $\delta_0 > 0$  such that*

$$\forall x \in A, 0 < |x - c| < \delta_0 \Rightarrow f(x) \leq h(x) \leq g(x).$$

*Then  $\lim_{x \rightarrow c} h(x) = L$ .*

*Go to proof.*

**Theorem 0.5.13** (Limits Are Local). *Let  $E \subseteq X \subseteq \mathbb{R}$ , let  $x_0$  be a cluster point of  $E$ , let  $f : X \rightarrow \mathbb{R}$ , and let  $\delta > 0$ . Then*

$$\lim_{x \rightarrow x_0; x \in E} f(x) = L \iff \lim_{x \rightarrow x_0; x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

*Go to proof.*

**Proposition 0.5.14** (Limit of Absolute Value). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$ , then  $\lim_{x \rightarrow c} |f(x)| = |L|$ .*

*Go to proof.*

**Theorem 0.5.15** (Composition of Limits). *Let  $f : A \rightarrow B$ , let  $g : B \rightarrow \mathbb{R}$ , let  $c_1$  be a cluster point of  $A$ , and let  $c_2 \in \mathbb{R}$  be a cluster point of  $B$ . Suppose  $\lim_{x \rightarrow c_1} f(x) = c_2$  and  $\lim_{y \rightarrow c_2} g(y) = L$ . If  $c_2 \in B$ , assume also that  $g(c_2) = L$ . Then  $\lim_{x \rightarrow c_1} g(f(x)) = L$ .*

*Go to proof.*

**Theorem 0.5.16** (Monotone Limit Theorem). *Let  $a < b$  and let  $f : (a, b) \rightarrow \mathbb{R}$  be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing  $f$  and every  $x_0 \in (a, b)$ :*

$$f(x_0^-) = \sup\{f(x) : a < x < x_0\}, \quad f(x_0^+) = \inf\{f(x) : x_0 < x < b\}.$$

*Go to proof.*

**Proposition 0.5.17** (Cauchy Criterion for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  exists for some  $L \in \mathbb{R}$  if and only if*

$$\forall \text{OutputTolerance}(\varepsilon), \exists \text{InputTolerance}(\delta), \forall x, y \in A : 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon. \blacksquare$$

*Go to proof.*

## Algebra of Limits

**Theorem 0.5.18** (Sum Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then*

$$\lim_{x \rightarrow c} (f + g)(x) = L + M.$$

*Go to proof.*



**Theorem 0.5.19** (Difference Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then*

$$\lim_{x \rightarrow c} (f - g)(x) = L - M.$$

*Go to proof.*

**Theorem 0.5.20** (Scalar Multiple Rule for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ , and let  $\alpha \in \mathbb{R}$ . If  $\lim_{x \rightarrow c} f(x) = L$ , then*

$$\lim_{x \rightarrow c} (\alpha f)(x) = \alpha L.$$

*Go to proof.*

**Theorem 0.5.21** (Product Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then*

$$\lim_{x \rightarrow c} (fg)(x) = LM.$$

*Go to proof.*

**Theorem 0.5.22** (Quotient Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$ ,  $\lim_{x \rightarrow c} g(x) = M$ , and  $M \neq 0$ , then there exists  $\delta_0 > 0$  such that  $g(x) \neq 0$  whenever  $x \in A$  and  $0 < |x - c| < \delta_0$ , and*

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

*Go to proof.*

## One-Sided Limits

**Definition 0.5.23** (Right-Hand Limit). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (c, \infty)$ . A real number  $L$  is the **right-hand limit** of  $f$  at  $c$  if*

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

**Definition 0.5.24** (Left-Hand Limit). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (-\infty, c)$ . A real number  $L$  is the **left-hand limit** of  $f$  at  $c$  if*

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

**Theorem 0.5.25** (Sequential Criterion for Right-Hand Limits). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (c, \infty)$ . Then  $\lim_{x \rightarrow c^+} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \cap (c, \infty)$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Go to proof.*

**Corollary 0.5.26** (Sequential Criterion for Left-Hand Limits). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (-\infty, c)$ . Then  $\lim_{x \rightarrow c^-} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \cap (-\infty, c)$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Go to proof.*

**Theorem 0.5.27** (Two-Sided Limit via One-Sided Limits). *Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . If  $c$  is a cluster point of both  $A \cap (c, \infty)$  and  $A \cap (-\infty, c)$ , then  $\lim_{x \rightarrow c} f(x) = L$  if and only if both  $\lim_{x \rightarrow c^+} f(x) = L$  and  $\lim_{x \rightarrow c^-} f(x) = L$ .*

*Go to proof.*

## 0.6 Point Continuity

### Continuity

**Definition 0.6.1** (Relative Neighborhood). Let  $E \subseteq \mathbb{R}$  and  $a \in E$ . A set  $U \subseteq \mathbb{R}$  is a neighborhood of  $a$  in  $E$  if and only if there exists  $\delta > 0$  such that  $U = E \cap (a - \delta, a + \delta)$ . Equivalently, every neighborhood of  $a$  in  $E$  is a set of the form  $U_E(a) := E \cap (a - \delta, a + \delta)$  for some  $\delta > 0$ .

**Definition 0.6.2** (Continuity at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is **continuous at  $c$**  if and only if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A : (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon).$$

**Definition 0.6.3** (Continuity at a Point — Neighbourhood Form). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is continuous at  $c$  if and only if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$f(A \cap (c - \delta, c + \delta)) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

**Definition 0.6.4** (Continuity at a Point — Sequential Form). Let  $A \subseteq \mathbb{R}$ , let  $c \in A$ , and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is continuous at  $c$  if and only if for every sequence  $(x_n)_{n \in \mathbb{N}}$  in  $A$ , the convergence  $\lim_{n \rightarrow \infty} x_n = c$  implies  $\lim_{n \rightarrow \infty} f(x_n) = f(c)$ .

### Characterization of Discontinuity

**Definition 0.6.5** (Point of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $a \in A$ . The function  $f$  is discontinuous at  $a$  if and only if  $f$  is not continuous at  $a$ .

**Definition 0.6.6** (Sequential Characterization of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is discontinuous at  $c$  if and only if there exists a sequence  $(x_n)_{n \in \mathbb{N}} \subseteq A$  such that  $x_n \rightarrow c$  while  $f(x_n) \not\rightarrow f(c)$ .

**Definition 0.6.7** (Neighbourhood Characterization of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is discontinuous at  $c$  if and only if there exists a neighborhood  $V$  of  $f(c)$  such that for every neighborhood  $U$  of  $c$  relative to  $A$  there exists  $x \in U \cap A$  with  $f(x) \notin V$ .

**Lemma 0.6.8** (Equivalent Formulations of Discontinuity at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The following are equivalent:

1.  $c$  is a point of discontinuity in the  $\varepsilon$ - $\delta$  sense.
2.  $c$  is a sequential point of discontinuity.
3.  $c$  is a neighbourhood point of discontinuity.

*Go to proof.*

**Definition 0.6.9** (Types of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $x_0 \in A$  with  $f$  discontinuous at  $x_0$ .

- (i) *Removable* iff  $\exists L \in \mathbb{R} : \lim_{x \rightarrow x_0} f(x) = L$  and  $L \neq f(x_0)$ .
- (ii) *Jump* iff  $\exists L_-, L_+ \in \mathbb{R} : \lim_{x \rightarrow x_0^-} f(x) = L_-, \lim_{x \rightarrow x_0^+} f(x) = L_+,$  and  $L_- \neq L_+.$
- (iii) *Essential* iff it is not removable.

## Oscillation

**Definition 0.6.10** (Oscillation on a Set). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $E \subseteq A$ . The oscillation of  $f$  on  $E$  is the extended real number

$$\omega(f; E) := \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\} = \text{diam}(f(E)).$$

**Definition 0.6.11** (Oscillation at a Point). Let  $E \subseteq \mathbb{R}$ , let  $f : E \rightarrow \mathbb{R}$ , and let  $x_0 \in \mathbb{R}$  satisfy  $U_\delta(x_0) \cap E \neq \emptyset$  for every  $\delta > 0$ , where  $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$ . The oscillation of  $f$  at  $x_0$  is the extended real number

$$\omega(f; x_0) = \lim_{\delta \rightarrow 0^+} \omega(f; U_\delta(x_0) \cap E),$$

with  $\omega(f; A)$  denoting the oscillation of  $f$  on the set  $A$ . Because the function  $\delta \mapsto \omega(f; U_\delta(x_0) \cap E)$  is nonincreasing, this right-hand limit exists (possibly  $+\infty$ ).

**Theorem 0.6.12** (Continuity Characterised by Oscillation). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . Then  $f$  is continuous at  $c$  if and only if  $\omega(f; c) = 0$ . Go to proof.

**Proposition 0.6.13** (Discontinuity Set via Oscillation). Let  $f : E \rightarrow \mathbb{R}$ . Then

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

Go to proof.

## 0.7 Global Theorems

### Intermediate Value Theorem

**Definition 0.7.1** (Bounded on a Set). Let  $A \subseteq \mathbb{R}$  and  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is bounded on  $A$  if and only if there exists a constant  $M > 0$  such that  $|f(x)| \leq M$  for every  $x \in A$ .

**Theorem 0.7.2** (Boundedness Theorem). Let  $a < b$  and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then there exists  $M > 0$  such that  $|f(x)| \leq M$  for every  $x \in [a, b]$ . Go to proof.

**Definition 0.7.3** (Absolute Maximum and Absolute Minimum). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . A point  $x^* \in A$  is an absolute maximum of  $f$  on  $A$  if and only if  $f(x^*) \geq f(x)$  for every  $x \in A$ . A point  $x_* \in A$  is an absolute minimum of  $f$  on  $A$  if and only if  $f(x_*) \leq f(x)$  for every  $x \in A$ .

**Theorem 0.7.4** (Extreme Value Theorem). Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. There exist points  $x_m, x_M \in [a, b]$  such that  $f(x_m) \leq f(x) \leq f(x_M)$  for every  $x \in [a, b]$ . Equivalently,  $f$  attains both an absolute maximum and an absolute minimum on  $[a, b]$ . Go to proof.

**Theorem 0.7.5** (Location of Roots). Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. If either  $f(a) < 0 < f(b)$  or  $f(a) > 0 > f(b)$ , then there exists  $c \in (a, b)$  such that  $f(c) = 0$ . Go to proof.

**Theorem 0.7.6** (Bolzano's Intermediate Value Theorem). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and choose  $a, b \in I$  with  $f(a) < k < f(b)$ . Then there exists  $c \in I$  such that  $\min\{a, b\} < c < \max\{a, b\}$  and  $f(c) = k$ . Go to proof.

**Theorem 0.7.7** (Image of Closed Bounded Interval Is Closed Bounded Interval). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. Then  $f([a, b]) = [\min_{[a, b]} f, \max_{[a, b]} f]$ . Go to proof.

**Theorem 0.7.8** (Preservation of Intervals). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be continuous. Then  $f(I)$  is an interval. Go to proof.

**Theorem 0.7.9** (Darboux Property). Let  $a, b \in \mathbb{R}$  satisfy  $a < b$ , let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and fix  $c \in \mathbb{R}$ . If  $f(x) \neq c$  for every  $x \in [a, b]$ , then either  $f(x) > c$  for all  $x \in [a, b]$  or  $f(x) < c$  for all  $x \in [a, b]$ . Go to proof.

## 0.8 Uniform Continuity

### Uniform Continuity

**Definition 0.8.1** (Uniform Continuity). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is uniformly continuous on  $A$  if and only if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that for all  $x, u \in A$  the implication  $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$  holds.

**Theorem 0.8.2** (Algebra of Uniform Continuity on a General Domain). (a) For every set  $A \subseteq \mathbb{R}$ , every pair of functions  $f, g : A \rightarrow \mathbb{R}$  that are uniformly continuous on  $A$ , and every scalar  $c \in \mathbb{R}$ , the functions  $f + g$ ,  $f - g$ , and  $cf$  are uniformly continuous on  $A$ .

(b) There exist a set  $A \subseteq \mathbb{R}$  and uniformly continuous functions  $f, g : A \rightarrow \mathbb{R}$  such that the product  $fg$  is not uniformly continuous on  $A$ .

Go to proof.

**Theorem 0.8.3** (Algebra of Uniform Continuity on a Bounded Domain). Let  $A \subseteq \mathbb{R}$  be bounded, and let  $f, g : A \rightarrow \mathbb{R}$  be uniformly continuous on  $A$ .

(a) The product  $x \mapsto f(x)g(x)$  is uniformly continuous on  $A$ .

(b) If there exists  $k > 0$  such that  $|g(x)| \geq k$  for all  $x \in A$ , then the quotient  $x \mapsto f(x)/g(x)$  is uniformly continuous on  $A$ .

Go to proof.

**Proposition 0.8.4** (Sequential Characterization of Uniform Continuity). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . Then  $f$  is uniformly continuous on  $A$  if and only if for every pair of sequences  $(x_n)$  and  $(y_n)$  in  $A$  with  $\lim_{n \rightarrow \infty} (x_n - y_n) = 0$  we have  $\lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) = 0$ . Go to proof.

**Theorem 0.8.5** (Heine–Cantor Theorem). Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then  $f$  is uniformly continuous on  $[a, b]$ . Go to proof.

**Proposition 0.8.6** (Uniform Continuity Preserves Cauchy Sequences). Let  $S \subseteq \mathbb{R}$ , let  $f : S \rightarrow \mathbb{R}$ , and let  $(x_n)$  be a sequence in  $S$ . If  $f$  is uniformly continuous on  $S$  and  $(x_n)$  is a Cauchy sequence in  $S$ , then  $(f(x_n))$  is a Cauchy sequence in  $\mathbb{R}$ . Go to proof.

**Definition 0.8.7** (Lipschitz Condition). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $K > 0$ . The function  $f$  satisfies the Lipschitz condition on  $A$  with constant  $K$  if and only if  $|f(x) - f(u)| \leq K|x - u|$  for every  $x, u \in A$ .

**Proposition 0.8.8** (Lipschitz Implies Uniform Continuity). Let  $A \subseteq \mathbb{R}$  and  $f : A \rightarrow \mathbb{R}$ . If there exists  $K \geq 0$  such that  $|f(x) - f(y)| \leq K|x - y|$  for all  $x, y \in A$ , then  $f$  is uniformly continuous on  $A$ . Go to proof.

**Theorem 0.8.9** (Algebra of Lipschitz Functions). Let  $A \subseteq \mathbb{R}$  and let  $f, g : A \rightarrow \mathbb{R}$  be Lipschitz with constants  $K_f, K_g \geq 0$  respectively.

(a) For every  $c \in \mathbb{R}$ , the function  $cf$  is Lipschitz on  $A$  with constant  $|c|K_f$ .

(b) The functions  $f + g$  and  $f - g$  are Lipschitz on  $A$  with constant  $K_f + K_g$ .

(c) If  $B \subseteq \mathbb{R}$ ,  $h : B \rightarrow \mathbb{R}$  is Lipschitz with constant  $K_h \geq 0$ , and  $f(A) \subseteq B$ , then  $h \circ f$  is Lipschitz on  $A$  with constant  $K_h K_f$ .

(d) If  $|f(x)| \leq M_f$  and  $|g(x)| \leq M_g$  for all  $x \in A$ , then  $fg$  is Lipschitz on  $A$  with constant  $M_g K_f + M_f K_g$ .

(e) If  $|f(x)| \leq M_f$  for all  $x \in A$ , if  $|g(x)| \geq k > 0$  for all  $x \in A$ , and if  $g$  is Lipschitz on  $A$  with constant  $K_g$ , then  $f/g$  is Lipschitz on  $A$  with constant  $K_f/k + M_f K_g/k^2$ .

Go to proof.

**Definition 0.8.10** (Bi-Lipschitz Map). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The map  $f$  is bi-Lipschitz on  $A$  if and only if there exist constants  $0 < \alpha \leq K$  such that for every  $x, u \in A$ ,

$$\alpha|x - u| \leq |f(x) - f(u)| \leq K|x - u|.$$

Any such pair  $(\alpha, K)$  is called a bi-Lipschitz constant pair for  $f$  on  $A$ .

**Theorem 0.8.11** (Inverse of a Bi-Lipschitz Map Is Lipschitz). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose there exist constants  $\alpha, K > 0$  such that  $\alpha|x - y| \leq |f(x) - f(y)| \leq K|x - y|$  for all  $x, y \in A$ . Then  $f$  is injective, the inverse mapping  $f^{-1} : f(A) \rightarrow A$  is well defined, and

$$\frac{1}{K}|u - v| \leq |f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{\alpha}|u - v| \quad \text{for all } u, v \in f(A).$$

Go to proof.

## 0.9 Monotone Functions

### Monotone Functions and Continuity

**Theorem 0.9.1** (One-Sided Limits of Monotone Functions). Let  $I \subseteq \mathbb{R}$  be an interval, let  $c$  be interior to  $I$ , and let  $f : I \rightarrow \mathbb{R}$  be increasing. Define

$$L_- := \sup\{f(x) : x \in I, x < c\}, \quad L_+ := \inf\{f(x) : x \in I, c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \rightarrow c^-} f(x) = L_-, \quad \lim_{x \rightarrow c^+} f(x) = L_+.$$

Go to proof.

**Corollary 0.9.2** (Continuity Criterion for Monotone Functions). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be monotone increasing, and let  $c$  be an interior point of  $I$ . Then  $f$  is continuous at  $c$  if and only if its one-sided limits satisfy  $f(c^-) = f(c) = f(c^+)$ , where  $f(c^-) = \lim_{x \rightarrow c^-} f(x)$  and  $f(c^+) = \lim_{x \rightarrow c^+} f(x)$ . Go to proof.

**Definition 0.9.3** (Jump of a Function). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be increasing, and let  $c$  be an interior point of  $I$ . If the one-sided limits  $f(c^-)$  and  $f(c^+)$  exist, the jump of  $f$  at  $c$  is

$$j_f(c) := f(c^+) - f(c^-).$$

**Proposition 0.9.4** (Discontinuities of a Monotone Function Are of First Kind). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be monotone, and let  $a \in I$  be such that there exist points of  $I$  strictly less and strictly greater than  $a$ . If  $f$  is discontinuous at  $a$ , then both one-sided limits  $\lim_{x \uparrow a} f(x)$  and  $\lim_{x \downarrow a} f(x)$  exist in  $\mathbb{R}$ . Go to proof.

**Corollary 0.9.5** (Jump Intervals Are Disjoint). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be nondecreasing. If  $a, b \in I$  are distinct points of discontinuity of  $f$ , then the open intervals  $(f(a^-), f(a^+))$  and  $(f(b^-), f(b^+))$  are disjoint. Go to proof.

**Theorem 0.9.6** (Discontinuities of a Monotone Function Are Countable). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be monotone. Then the set  $D_f := \{c \in I : f \text{ is discontinuous at } c\}$  is at most countable. Go to proof.

**Corollary 0.9.7** (Monotone Function Is Continuous on a Dense Set). Let  $f : [a, b] \rightarrow \mathbb{R}$  be monotone. Then the set  $\{x \in [a, b] : f \text{ is continuous at } x\}$  is dense in  $[a, b]$ . Go to proof.

**Proposition 0.9.8** (Continuous Injective Is Strictly Monotone). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then  $f$  is injective if and only if  $f$  is strictly monotone on  $[a, b]$ . Go to proof.

**Theorem 0.9.9** (Continuous Inverse Theorem). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be strictly monotone and continuous on  $I$ . Then the inverse function  $f^{-1} : f(I) \rightarrow I$  exists, is strictly monotone, and is continuous on  $f(I)$ . Go to proof.

## Limits Superior and Inferior of Functions

**Definition 0.9.10** (Limits Superior and Inferior of a Function). Let  $f : E \rightarrow \mathbb{R}$  and let  $x_0$  be an accumulation point of  $E$ . For each  $\delta > 0$  set

$$U(\delta) = \{ f(x) : x \in E, 0 < |x - x_0| < \delta \}.$$

The extended real numbers  $\limsup_{x \rightarrow x_0} f(x)$  and  $\liminf_{x \rightarrow x_0} f(x)$  are defined by

$$\limsup_{x \rightarrow x_0} f(x) = \inf_{\delta > 0} \sup U(\delta), \quad \liminf_{x \rightarrow x_0} f(x) = \sup_{\delta > 0} \inf U(\delta),$$

where each inner supremum or infimum is taken in  $\overline{\mathbb{R}}$  and the outer infimum or supremum ranges over all  $\delta > 0$ .

**Proposition 0.9.11** ( $\limsup \geq \liminf$ ). Let  $A \subseteq \mathbb{R}$ , let  $x_0$  be a limit point of  $A$ , and let  $f : A \rightarrow \mathbb{R}$ . Then

$$\limsup_{x \rightarrow x_0} f(x) \geq \liminf_{x \rightarrow x_0} f(x).$$

*Go to proof.*

**Proposition 0.9.12** (Limit Exists iff  $\limsup = \liminf$ ). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $x_0$  be a limit point of  $A$  with  $x_0 \in \mathbb{R}$ , and let  $L \in \mathbb{R}$ . Then  $\lim_{x \rightarrow x_0} f(x)$  exists and equals  $L$  if and only if  $\limsup_{x \rightarrow x_0} f(x) = \liminf_{x \rightarrow x_0} f(x) = L$ . *Go to proof.*

## 0.10 Limits at Infinity

### Limits at Infinity

**Definition 0.10.1** (Infinite Adherent Points). Let  $X \subseteq \mathbb{R}$ . The point  $+\infty$  is adherent to  $X$  if and only if for every  $M \in \mathbb{R}$  there exists  $x \in X$  such that  $x > M$ . The point  $-\infty$  is adherent to  $X$  if and only if for every  $M \in \mathbb{R}$  there exists  $x \in X$  such that  $x < M$ .

**Definition 0.10.2** (Limit at Infinity). Let  $X \subseteq \mathbb{R}$  be such that for every  $M \in \mathbb{R}$  there exists  $x \in X$  with  $x > M$ , let  $f : X \rightarrow \mathbb{R}$ , and let  $L \in \mathbb{R}$ . The limit of  $f$  at  $+\infty$  equals  $L$  if and only if for every  $\varepsilon > 0$  there exists  $M \in \mathbb{R}$  such that for every  $x \in X$ ,  $x > M$  implies  $|f(x) - L| < \varepsilon$ .

## 0.11 Approximation

### Approximation of Continuous Functions

**Theorem 0.11.1** (Step Function Approximation). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ . Then there exists a step function  $s_\varepsilon : [a, b] \rightarrow \mathbb{R}$  such that

$$\sup_{x \in [a, b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

*Go to proof.*

**Definition 0.11.2** (Piecewise Linear Function). Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $g : [a, b] \rightarrow \mathbb{R}$ . The function  $g$  is piecewise linear if and only if there exist  $m \in \mathbb{N}$  with  $m \geq 1$  and points  $a = x_0 < x_1 < \dots < x_m = b$  such that  $g|_{[x_{k-1}, x_k]}$  is affine for every  $k \in \{1, \dots, m\}$ .

**Theorem 0.11.3** (Piecewise Linear Approximation). Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ . Then there exists a continuous piecewise linear function  $\ell_\varepsilon : [a, b] \rightarrow \mathbb{R}$  such that  $\sup_{x \in [a, b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$ . *Go to proof.*

**Theorem 0.11.4** (Weierstrass Approximation Theorem). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. For every  $\varepsilon > 0$  there exists a polynomial  $p_\varepsilon \in \mathbb{R}[x]$  such that  $\|f - p_\varepsilon\|_{\infty, [a, b]} < \varepsilon$ . Go to proof.*

**Definition 0.11.5** (Bernstein Polynomial). Let  $n \in \mathbb{N}$  and let  $f : [0, 1] \rightarrow \mathbb{R}$ . A function  $B_n : [0, 1] \rightarrow \mathbb{R}$  is called the  $n$ th Bernstein polynomial of  $f$  if and only if, for every  $x \in [0, 1]$ ,

$$B_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

**Theorem 0.11.6** (Bernstein Approximation Theorem). *Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous, and for each  $n \in \mathbb{N}$  let  $B_n$  denote the  $n$ th Bernstein polynomial of  $f$ . Then for every  $\varepsilon > 0$  there exists  $n_\varepsilon \in \mathbb{N}$  such that  $\|f - B_n\|_\infty < \varepsilon$  for all  $n \geq n_\varepsilon$ . Go to proof.*

## 0.12 Gauge

### Gauges and Tagged Partitions

**Definition 0.12.1** (Partition). Let  $a < b$  and set  $I = [a, b]$ . A finite ordered set  $P = \{x_0, \dots, x_n\}$  with  $n \in \mathbb{N}$  and  $n \geq 1$  is a partition of  $I$  if and only if  $x_0 = a$ ,  $x_n = b$ , and  $x_0 < x_1 < \dots < x_n$ . In this case the closed subintervals  $[x_{i-1}, x_i]$  for  $i = 1, \dots, n$  subdivide  $I$ .

**Definition 0.12.2** (Tagged Partition). Let  $a, b \in \mathbb{R}$  with  $a < b$ . A list  $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  is a tagged partition of  $[a, b]$  if and only if the subintervals determine a partition  $P = \{[x_{i-1}, x_i]\}_{i=1}^n$  of  $[a, b]$  with  $a = x_0 < x_1 < \dots < x_n = b$ , and each tag satisfies  $t_i \in [x_{i-1}, x_i]$  for  $i = 1, \dots, n$ . Equivalently, a tagged partition is exactly a partition  $P$  of  $[a, b]$  together with one tag chosen inside every subinterval of  $P$ .

**Definition 0.12.3** (Gauge). Let  $a, b \in \mathbb{R}$  with  $a < b$  and set  $I = [a, b]$ . A gauge on  $I$  is a function  $\delta : I \rightarrow (0, \infty)$ .

**Definition 0.12.4** ( $\delta$ -Fine Partition). Let  $[a, b]$  be a closed interval, let  $\delta : [a, b] \rightarrow (0, \infty)$  be a gauge, and let  $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  be a tagged partition of  $[a, b]$ . The tagged partition  $\dot{P}$  is  $\delta$ -fine if and only if for every  $i \in \{1, \dots, n\}$  and every  $x \in [x_{i-1}, x_i]$  the inequality  $|x - t_i| < \delta(t_i)$  holds.

**Lemma 0.12.5** (Every Point Is Covered by a Tag). *Let  $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  be a tagged partition of  $[a, b]$  that is  $\delta$ -fine on  $[a, b]$ , and let  $x \in [a, b]$ . Then there exists  $i \in \{1, \dots, n\}$  such that  $|x - t_i| < \delta(t_i)$ . Go to proof.*

**Theorem 0.12.6** (Cousin's Theorem). *Every gauge  $\delta : [a, b] \rightarrow (0, \infty)$  on a closed interval  $[a, b]$  admits a  $\delta$ -fine tagged partition of  $[a, b]$ . Go to proof.*

### Tagged Partitions and Mesh

**Definition 0.12.7** (Mesh of a Partition). Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $P = \{x_0, \dots, x_n\}$  be a partition of  $[a, b]$ . A real number  $m$  is the *mesh* (or *norm*) of  $P$  if and only if

$$m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

The mesh of  $P$  is denoted by  $\|P\|$  and is the unique  $m$  satisfying this equivalence.

**Definition 0.12.8** (Refinement). Let  $[a, b] \subseteq \mathbb{R}$  and let  $P$  and  $Q$  be partitions of  $[a, b]$ . Then  $Q$  is a refinement of  $P$  if and only if every point of  $P$  is also a point of  $Q$ ; equivalently,  $P \subseteq Q$ .

**Definition 0.12.9** (Common Refinement). Let  $a < b$  and let  $P$  and  $Q$  be partitions of  $[a, b]$ . Their common refinement is the partition  $R$  obtained by taking the union of their node sets, so  $R = P \cup Q$  as sets of points in  $[a, b]$ .

## Proofs

# Differentiation

## 0.13 The Tangent Problem

**Definition 0.13.1** (Secant Line). Let  $f : A \rightarrow \mathbb{R}$ , and let  $x_1, x_2 \in A$  with  $x_1 \neq x_2$ . The *secant line* determined by the points  $(x_1, f(x_1))$  and  $(x_2, f(x_2))$  is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

**Definition 0.13.2** (Difference Quotient). Let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$ , and let  $x \in A$  with  $x \neq c$ . The *difference quotient* of  $f$  from  $c$  to  $x$  is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

**Definition 0.13.3** (Tangent Line). Let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$ , and suppose that  $f$  is differentiable at  $c$ . The *tangent line* to the graph of  $f$  at the point  $(c, f(c))$  is the line through  $(c, f(c))$  with slope  $f'(c)$ . Its equation is

$$y = f(c) + f'(c)(x - c).$$

## 0.14 The Derivative

**Definition 0.14.1** (Derivative at a Point). Let  $f : D \rightarrow \mathbb{R}$  be a real-valued function and  $c \in \text{int}(D)$ . We say  $f$  is *differentiable at  $c$*  with *derivative  $L = f'(c)$*  if and only if:

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D \left( 0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

**Definition 0.14.2** (Topological Definition of Derivative at a Point). Let  $f : D \rightarrow \mathbb{R}$  and  $c \in \text{int}(D)$ .  $f$  is differentiable at  $c$  with derivative  $L$  if for every neighbourhood  $V$  of  $L$ , there exists a neighbourhood  $U$  of  $c$  such that for all  $x \in (U \setminus \{c\}) \cap D$ :

$$\frac{f(x) - f(c)}{x - c} \in V.$$

**Definition 0.14.3** (Sequential Definition of Derivative at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . We say that  $f$  is differentiable at  $c$  with derivative  $L$  if

$$\forall (x_n) \subseteq A \setminus \{c\} \left( x_n \rightarrow c \implies \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L \right).$$

**Theorem 0.14.4** (Equivalence of Definitions of the Derivative at a Point). Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$  be a limit point of  $A$ . The following are equivalent:

1.  $f$  is differentiable at  $c$  in the  $\varepsilon$ - $\delta$  sense with derivative  $L$ .
2.  $f$  is differentiable at  $c$  in the topological sense with derivative  $L$ .
3.  $f$  is differentiable at  $c$  in the sequential sense with derivative  $L$ .



Go to proof.

**Proposition 0.14.5** (Derivative:  $h$ -Form Equivalence). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in \text{int}(A)$ . Then  $f$  is differentiable at  $c$  with derivative  $L$  if and only if*

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

### 0.14.1 First Consequences

**Theorem 0.14.6** (Differentiable Implies Continuous). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$ . If  $f$  is differentiable at  $c$ , then  $f$  is continuous at  $c$ .*

Go to proof.

**Theorem 0.14.7** (Uniqueness of the Derivative). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$ . If  $f$  is differentiable at  $c$ , its derivative is unique.*

Go to proof. Go to proof.

**Proposition 0.14.8** (Derivative of the Identity). *Let  $f(x) = x$  for all  $x \in \mathbb{R}$ . Then  $f'(c) = 1$  for every  $c \in \mathbb{R}$ .*

Go to proof.

**Proposition 0.14.9** (Derivative of a Constant). *Let  $k \in \mathbb{R}$  and let  $f(x) = k$  for all  $x \in \mathbb{R}$ . Then  $f'(c) = 0$  for every  $c \in \mathbb{R}$ .*

Go to proof.

## 0.15 One-Sided Derivatives

**Definition 0.15.1** (Left-Hand Derivative). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$  be such that  $c$  is a left-hand limit point of  $I$ . We say that  $f$  has a *left-hand derivative* at  $c$  if the limit

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted  $f'_-(c)$ :

$$f'_-(c) := \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}.$$

**Definition 0.15.2** (Right-Hand Derivative). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$  be such that  $c$  is a right-hand limit point of  $I$ . We say that  $f$  has a *right-hand derivative* at  $c$  if the limit

$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted  $f'_+(c)$ :

$$f'_+(c) := \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}.$$

**Theorem 0.15.3** (Differentiability and One-Sided Derivatives). *Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f : I \rightarrow \mathbb{R}$ . Then  $f$  is differentiable at  $c \in I$  if and only if both one-sided derivatives  $f'_-(c)$  and  $f'_+(c)$  exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

## 0.16 Carathéodory and the Chain Rule

### 0.16.1 Carathéodory Characterization

**Definition 0.16.1** (Carathéodory Factorization). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . We say that  $f$  admits a *Carathéodory factorization* at  $c$  if there exists a function  $\phi : I \rightarrow \mathbb{R}$ , continuous at  $c$ , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case,  $f$  is differentiable at  $c$  and  $f'(c) = \phi(c)$ .

**Theorem 0.16.2** (Carathéodory Characterization of Differentiability). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . The following are equivalent:

- (i)  $f$  is differentiable at  $c$ .
- (ii) There exists a function  $\phi : I \rightarrow \mathbb{R}$ , continuous at  $c$ , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case,  $\phi(c) = f'(c)$ . Go to proof. Go to proof.

**Theorem 0.16.3** (Chain Rule). Let  $I, J \subseteq \mathbb{R}$  be intervals, let  $f : I \rightarrow \mathbb{R}$  and  $g : J \rightarrow \mathbb{R}$ , and suppose that  $f(I) \subseteq J$ . Let  $c \in I$ . If  $f$  is differentiable at  $c$  and  $g$  is differentiable at  $f(c)$ , then the composition  $g \circ f$  is differentiable at  $c$ , and

$$(g \circ f)'(c) = g'(f(c)) f'(c).$$

Go to proof.

**Theorem 0.16.4** (Leibniz Rule). Let  $I \subseteq \mathbb{R}$  be an interval, let  $x \in I$ , and suppose that  $f, g : I \rightarrow \mathbb{R}$  are  $n$ -times differentiable on a neighbourhood of  $x$ . Then  $fg$  is  $n$ -times differentiable at  $x$ , and

$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Go to proof.

**Theorem 0.16.5** (Faa di Bruno's Formula). Let  $I, J \subseteq \mathbb{R}$  be intervals, let  $x \in I$ , suppose  $g : I \rightarrow J$  is  $n$ -times differentiable on a neighbourhood of  $x$ , and suppose  $f : J \rightarrow \mathbb{R}$  is  $n$ -times differentiable on a neighbourhood of  $g(x)$ . Then  $f \circ g$  is  $n$ -times differentiable at  $x$ , and

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^n m k_m = n} \frac{n!}{\prod_{m=1}^n k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^n (g^{(m)}(x))^{k_m},$$

where  $k_m \geq 0$  and  $k = \sum_{m=1}^n k_m$ . Go to proof.

## 0.17 The Shape Questions

**Definition 0.17.1** (Convex Function). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *convex* on  $I$  if for all  $x, y \in I$  and all  $\lambda \in [0, 1]$ ,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

**Definition 0.17.2** (Concave Function). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *concave* on  $I$  if  $-f$  is convex on  $I$ .

**Definition 0.17.3** (Inflection Point). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$  be an interior point. The point  $c$  is an *inflection point* of  $f$  if  $f$  changes convexity at  $c$ : there is a neighbourhood  $(c - \delta, c + \delta) \subseteq I$  such that  $f$  is convex on one side of  $c$  and concave on the other side.

### 0.17.1 Interior Extrema

**Definition 0.17.4** (Relative Minimum). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  has a *relative minimum* at  $c$  if there exists  $\delta > 0$  such that

$$f(c) \leq f(x) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(c) < f(x) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then  $f$  has a *strict relative minimum* at  $c$ .

**Definition 0.17.5** (Relative Maximum). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  has a *relative maximum* at  $c$  if there exists  $\delta > 0$  such that

$$f(x) \leq f(c) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(x) < f(c) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then  $f$  has a *strict relative maximum* at  $c$ .

**Definition 0.17.6** (Relative Extremum). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  has a *relative extremum* at  $c$  if  $f$  has either a relative minimum at  $c$  or a relative maximum at  $c$ .

**Theorem 0.17.7** (Interior Extremum Theorem). Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$  and  $f$  is differentiable at  $c$ , then

$$f'(c) = 0.$$

*Go to proof.*

**Theorem 0.17.8** (Necessary Condition for an Interior Extremum). Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$ , then either  $f'(c)$  does not exist, or

$$f'(c) = 0.$$

*Go to proof.*

**Corollary 0.17.9** (Necessary Condition for an Extremum). Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$ , then

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

*Go to proof.*

## 0.18 The Mean Value Theorems

**Theorem 0.18.1** (Rolle's Theorem). Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and satisfies  $f(a) = f(b)$ . Then there exists  $c \in (a, b)$  such that

$$f'(c) = 0.$$

*Go to proof.*

**Theorem 0.18.2** (Mean Value Theorem). Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Go to proof.*

**Theorem 0.18.3** (Cauchy Mean Value Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f, g : [a, b] \rightarrow \mathbb{R}$  are continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

*If, in addition,  $g'(x) \neq 0$  for every  $x \in (a, b)$ , then  $g(b) \neq g(a)$  and*

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

*Go to proof.*

**Corollary 0.18.4** (Derivative Bound Implies Lipschitz). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . If there exists  $M \geq 0$  such that*

$$|f'(x)| \leq M \quad \text{for every } x \in I,$$

*then  $f$  is Lipschitz on  $I$  with constant  $M$ :*

$$|f(x) - f(y)| \leq M|x - y| \quad \text{for every } x, y \in I.$$

*Go to proof.*

## 0.19 Reading the Graph from $f'$ and $f''$

### 0.19.1 First-Order Shape

**Definition 0.19.1** (Increasing at a Point). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  is increasing at  $c$  if there exists  $\delta > 0$  such that for all*

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

*we have  $f(x) \leq f(c) \leq f(y)$ .*

**Definition 0.19.2** (Decreasing at a Point). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  is decreasing at  $c$  if there exists  $\delta > 0$  such that for all*

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

*we have  $f(x) \geq f(c) \geq f(y)$ .*

**Theorem 0.19.3** (Zero Derivative Implies Constant). *Let  $I := [a, b] \subseteq \mathbb{R}$  be a closed interval, and let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$  and differentiable on  $(a, b)$ . If*

$$f'(x) = 0 \quad \text{for all } x \in (a, b),$$

*then  $f$  is constant on  $I$ . Go to proof.*

**Corollary 0.19.4** (Equal Derivatives Imply Difference Is Constant). *Let  $I := [a, b] \subseteq \mathbb{R}$ , and let  $f, g : I \rightarrow \mathbb{R}$  be continuous on  $I$  and differentiable on  $(a, b)$ . If*

$$f'(x) = g'(x) \quad \text{for all } x \in (a, b),$$

*then there exists  $C \in \mathbb{R}$  such that  $f(x) = g(x) + C$  for all  $x \in I$ . Go to proof.*

**Theorem 0.19.5** (Nondecreasing Iff Nonnegative Derivative). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Then  $f$  is nondecreasing on  $I$  if and only if*

$$f'(x) \geq 0 \quad \text{for all } x \in I.$$

*Go to proof.*

**Theorem 0.19.6** (Nonincreasing Iff Nonpositive Derivative). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Then  $f$  is nonincreasing on  $I$  if and only if*

$$f'(x) \leq 0 \quad \text{for all } x \in I.$$

*Go to proof.*

**Lemma 0.19.7** (Positive Derivative Implies Locally Increasing). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , let  $c \in I$ , and suppose that  $f$  is differentiable at  $c$ . If  $f'(c) > 0$ , then there exists  $\delta > 0$  such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

*and*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

*Go to proof.*

**Lemma 0.19.8** (Negative Derivative Implies Locally Decreasing). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , let  $c \in I$ , and suppose that  $f$  is differentiable at  $c$ . If  $f'(c) < 0$ , then there exists  $\delta > 0$  such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

*and*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

*Go to proof.*

**Theorem 0.19.9** (First Derivative Test: Relative Maximum). *Let  $I := [a, b] \subseteq \mathbb{R}$ , let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and let  $c \in (a, b)$ . Suppose  $f$  is differentiable on  $(a, c)$  and  $(c, b)$ . If there exists  $\delta > 0$  such that  $(c - \delta, c + \delta) \subseteq I$  and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \leq 0 \quad \text{for all } x \in (c, c + \delta),$$

*then  $f$  has a relative maximum at  $c$ . Go to proof.*

**Theorem 0.19.10** (First Derivative Test: Relative Minimum). *Let  $I := [a, b] \subseteq \mathbb{R}$ , let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and let  $c \in (a, b)$ . Suppose  $f$  is differentiable on  $(a, c)$  and  $(c, b)$ . If there exists  $\delta > 0$  such that  $(c - \delta, c + \delta) \subseteq I$  and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \geq 0 \quad \text{for all } x \in (c, c + \delta),$$

*then  $f$  has a relative minimum at  $c$ . Go to proof.*

## 0.19.2 Second-Order Shape

### 0.19.3 Higher-Order Derivatives

**Definition 0.19.11** (Second Derivative). *Let  $f$  be differentiable on a neighbourhood of  $c$ . If  $f'$  is differentiable at  $c$ , the *second derivative* of  $f$  at  $c$  is*

$$f''(c) := (f')'(c).$$

**Definition 0.19.12** (Higher Derivatives). *Define  $f^{(0)} := f$  and  $f^{(1)} := f'$ . For  $n \geq 2$ , if  $f^{(n-1)}$  is differentiable at  $c$ , define*

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

### 0.19.4 Second-Order Shape

**Theorem 0.19.13** (Second-Derivative Convexity Test). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be twice differentiable on  $I$ . If*

$$f''(x) \geq 0 \quad \text{for all } x \in I,$$

*then  $f$  is convex on  $I$ . Go to proof.*

**Theorem 0.19.14** (Second-Derivative Concavity Test). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be twice differentiable on  $I$ . If*

$$f''(x) \leq 0 \quad \text{for all } x \in I,$$

*then  $f$  is concave on  $I$ . Go to proof.*

**Theorem 0.19.15** (Second Derivative Test). *Let  $f$  be twice differentiable in a neighbourhood of  $c$ , and suppose  $f'(c) = 0$ .*

- *If  $f''(c) > 0$ , then  $f$  has a strict relative minimum at  $c$ .*
- *If  $f''(c) < 0$ , then  $f$  has a strict relative maximum at  $c$ .*

*Go to proof.*

**Proposition 0.19.16** (Inflection Point Necessary Condition). *Let  $f$  have an inflection point at  $c$ . If  $f''$  exists on a neighbourhood of  $c$  and is continuous at  $c$ , then*

$$f''(c) = 0.$$

*Go to proof.*

### 0.19.5 Darboux's Theorem

**Theorem 0.19.17** (Darboux's Theorem). *Let  $I := [a, b] \subseteq \mathbb{R}$  and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . If  $k$  is a real number strictly between  $f'(a)$  and  $f'(b)$ , then there exists  $c \in (a, b)$  such that*

$$f'(c) = k.$$

*Go to proof.*

### 0.19.6 Smoothness Classes

**Definition 0.19.18** (Continuously Differentiable; Class  $C^1$ ). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *continuously differentiable* on  $I$ , written  $f \in C^1(I)$ , if  $f$  is differentiable at every point of  $I$  and the derivative  $f' : I \rightarrow \mathbb{R}$  is continuous on  $I$ .*

**Definition 0.19.19** (Class  $C^k$ ). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $k \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is of *class  $C^k$*  on  $I$ , written  $f \in C^k(I)$ , if the derivatives  $f^{(0)}, f^{(1)}, \dots, f^{(k)}$  exist on  $I$  and  $f^{(k)}$  is continuous on  $I$ . By convention,  $C^0(I)$  is the class of continuous functions on  $I$ .*

**Definition 0.19.20** (Smooth Function; Class  $C^\infty$ ). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *smooth*, or of *class  $C^\infty$* , written  $f \in C^\infty(I)$ , if  $f \in C^k(I)$  for every  $k \in \mathbb{N}$ . Equivalently,*

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

**Definition 0.19.21** (Real-Analytic Function; Class  $C^\omega$ ). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *real-analytic*, or of *class  $C^\omega$* , written  $f \in C^\omega(I)$ , if for every  $a \in I$  there exists  $r > 0$  such that*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad \text{for every } x \in (a-r, a+r) \cap I.$$

**Theorem 0.19.22** (The Smoothness Tower). *For every nondegenerate interval  $I$ , the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

*hold, and each displayed inclusion is strict. Go to proof.*

**Definition 0.19.23** (Lipschitz-Derivative Class  $C^{1,1}$ ). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is of class  $C^{1,1}$  on  $I$ , written  $f \in C^{1,1}(I)$ , if  $f \in C^1(I)$  and  $f'$  is Lipschitz on  $I$ : there exists  $L \geq 0$  such that

$$|f'(x) - f'(y)| \leq L|x - y| \quad \text{for all } x, y \in I.$$

**Theorem 0.19.24** (Placement of  $C^{1,1}$ ). *For every nondegenerate interval  $I$ ,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

*and  $C^{1,1}(I)$  is not contained in  $C^2(I)$ . If  $K$  is a compact interval, then*

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

*and the first inclusion is strict. Go to proof.*

**Corollary 0.19.25** (Bounded Second Derivative Implies  $C^{1,1}$ ). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f \in C^2(I)$ . If  $|f''(x)| \leq M$  for all  $x \in I$ , then  $f \in C^{1,1}(I)$ , and  $f'$  is Lipschitz with constant  $M$ . Go to proof.*

## 0.20 The Algebra of Derivatives

### 0.20.1 From Characterization to Computation

**Theorem 0.20.1** (Constant Multiple Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$  be a limit point of  $A$ , and let  $\alpha \in \mathbb{R}$ . If  $f$  is differentiable at  $c$ , then  $\alpha f$  is differentiable at  $c$ , and*

$$(\alpha f)'(c) = \alpha f'(c).$$

*Go to proof.*

**Theorem 0.20.2** (Sum Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If  $f$  and  $g$  are differentiable at  $c$ , then  $f + g$  is differentiable at  $c$ , and*

$$(f + g)'(c) = f'(c) + g'(c).$$

*Go to proof.*

**Theorem 0.20.3** (Product Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If  $f$  and  $g$  are differentiable at  $c$ , then  $fg$  is differentiable at  $c$ , and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

*Go to proof.*

**Theorem 0.20.4** (Quotient Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . Suppose that  $f$  and  $g$  are differentiable at  $c$  and that  $g(c) \neq 0$ . Then  $f/g$  is differentiable at  $c$ , and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

*Go to proof.*

## 0.20.2 Closure

**Corollary 0.20.5** (Finite Sum Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then  $\sum_{i=1}^n \alpha_i f_i$  is differentiable at  $c$ , and*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

*Go to proof.*

**Corollary 0.20.6** (Extended Product Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then  $\prod_{i=1}^n f_i$  is differentiable at  $c$ , and*

$$\left( \prod_{i=1}^n f_i \right)'(c) = \sum_{k=1}^n f'_k(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

*Go to proof.*

**Corollary 0.20.7** (Power Rule for Integer Powers of a Function). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$  be a limit point of  $A$ , and let  $n \in \mathbb{N}$ . If  $f$  is differentiable at  $c$ , then  $f^n$  is differentiable at  $c$ , and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

*Go to proof.*

**Theorem 0.20.8** (Finite Linear Combination Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

*Go to proof.*

## 0.20.3 Global Forms on an Interval

**Theorem 0.20.9** (Constant Multiple Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $\alpha \in \mathbb{R}$ . If  $f$  is differentiable on  $I$ , then  $\alpha f$  is differentiable on  $I$ , and*

$$(\alpha f)'(x) = \alpha f'(x) \quad \text{for all } x \in I.$$

*Go to proof.*

**Theorem 0.20.10** (Sum Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . If  $f$  and  $g$  are differentiable on  $I$ , then  $f + g$  is differentiable on  $I$ , and*

$$(f + g)'(x) = f'(x) + g'(x) \quad \text{for all } x \in I.$$

*Go to proof.*

**Theorem 0.20.11** (Product Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . If  $f$  and  $g$  are differentiable on  $I$ , then  $fg$  is differentiable on  $I$ , and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \quad \text{for all } x \in I.$$

*Go to proof.*



**Theorem 0.20.12** (Quotient Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . Suppose that  $f$  and  $g$  are differentiable on  $I$  and that  $g(x) \neq 0$  for all  $x \in I$ . Then  $f/g$  is differentiable on  $I$ , and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad \text{for all } x \in I.$$

*Go to proof.*

**Corollary 0.20.13** (Finite Sum Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ , and let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then  $\sum_{i=1}^n \alpha_i f_i$  is differentiable on  $I$ , and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

*Go to proof.*

**Corollary 0.20.14** (Extended Product Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then  $\prod_{i=1}^n f_i$  is differentiable on  $I$ , and*

$$\left(\prod_{i=1}^n f_i\right)'(x) = \sum_{k=1}^n f'_k(x) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(x) \quad \text{for all } x \in I.$$

*Go to proof.*

**Corollary 0.20.15** (Power Rule for Integer Powers of a Function on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $n \in \mathbb{N}$ . If  $f$  is differentiable on  $I$ , then  $f^n$  is differentiable on  $I$ , and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \quad \text{for all } x \in I.$$

*Go to proof.*

**Theorem 0.20.16** (Finite Linear Combination Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ , and let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

*Go to proof.*

## 0.20.4 Inverting the Operation

**Theorem 0.20.17** (Inverse Function Theorem, One Variable). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$  be of class  $C^1$ , and let  $c \in I$  satisfy  $f'(c) \neq 0$ . Then there exist open intervals  $U \subseteq I$  and  $V \subseteq \mathbb{R}$ , with  $c \in U$  and  $f(c) \in V$ , such that:*

1.  $f|_U : U \rightarrow V$  is a bijection;
2. the inverse  $g = (f|_U)^{-1} : V \rightarrow U$  is of class  $C^1$ ;
3. for every  $y \in V$ ,

$$g'(y) = \frac{1}{f'(g(y))}.$$

*Go to proof.*

**Corollary 0.20.18** (Derivative of the Inverse Function). *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse  $g$  satisfies*

$$g' = \frac{1}{f' \circ g}$$

on  $V$ . More globally, if  $f : I \rightarrow J$  is a  $C^1$  bijection with  $f'(x) \neq 0$  for every  $x \in I$ , then  $f^{-1} : J \rightarrow I$  is  $C^1$  and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \quad \text{for every } y \in J.$$

Go to proof.

## 0.20.5 Indeterminate Forms: L'Hôpital's Rule

**Theorem 0.20.19** (L'Hôpital's Rule, 0/0 Form). *Let  $a < b$ , and let  $f, g : (a, b) \rightarrow \mathbb{R}$  be differentiable with  $g'(x) \neq 0$  for every  $x \in (a, b)$ . Suppose*

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

for  $L \in \mathbb{R} \cup \{-\infty, +\infty\}$ . Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

**Theorem 0.20.20** (L'Hôpital's Rule,  $\infty/\infty$  Form). *Let  $a < b$ , and let  $f, g : (a, b) \rightarrow \mathbb{R}$  be differentiable with  $g'(x) \neq 0$  for every  $x \in (a, b)$ . Suppose*

$$\lim_{x \rightarrow a^+} |g(x)| = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

for  $L \in \mathbb{R} \cup \{-\infty, +\infty\}$ . Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on  $\lim_{x \rightarrow a^+} f(x)$  is required. Go to proof.

## 0.21 Taylor: Construction and Its Error

**Definition 0.21.1** (Taylor Polynomial at a Point). Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have derivatives  $f^{(k)}(a)$  for  $0 \leq k \leq n$ . The **Taylor polynomial of order  $n$**  for  $f$  at  $a$  is

$$T_{n,a}f(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

**Definition 0.21.2** (Taylor Remainder). Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have a Taylor polynomial  $T_{n,a}f$ . The **Taylor remainder of order  $n$**  at  $a$  is the function  $R_{n,a}f : I \rightarrow \mathbb{R}$  defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

**Definition 0.21.3** (Maclaurin Polynomial). Let  $n \in \mathbb{N}$ , and let  $f$  be defined on an interval containing 0 with derivatives  $f^{(k)}(0)$  for  $0 \leq k \leq n$ . The **Maclaurin polynomial of order  $n$**  for  $f$  is

$$T_{n,0}f(x) := \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k.$$

**Theorem 0.21.4** (Taylor's Theorem with Lagrange Remainder). *Let  $a, b \in \mathbb{R}$  with  $a < b$ , let  $n \in \mathbb{N}$ , and let  $f : [a, b] \rightarrow \mathbb{R}$ . Suppose that  $f, f', \dots, f^{(n)}$  are continuous on  $[a, b]$  and that  $f^{(n+1)}$  exists on  $(a, b)$ . Then, for every  $x \in (a, b)$ , there exists  $c$  strictly between  $a$  and  $x$  such that*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

*Go to proof.*

**Corollary 0.21.5** (Taylor Expansion with Peano Remainder). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have derivatives up to order  $n$  in a neighbourhood of  $a$ . If  $f^{(n)}$  is continuous at  $a$ , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + o((x-a)^n) \quad \text{as } x \rightarrow a.$$

*Go to proof.*

**Corollary 0.21.6** (First-Order Peano Remainder). *Let  $f$  be differentiable at  $c$ . Then, as  $h \rightarrow 0$ ,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

*Go to proof.*

**Proposition 0.21.7** (The Flat Function: Smooth but Not Analytic). *Define  $f : \mathbb{R} \rightarrow \mathbb{R}$  by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

*Then:*

1.  $f \in C^\infty(\mathbb{R})$ ;
2.  $f^{(n)}(0) = 0$  for every  $n \geq 0$ ;
3. the Maclaurin series of  $f$  is identically 0, so  $f \notin C^\omega$  on any neighbourhood of 0.

*In particular,  $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \emptyset$ . Go to proof.*

## 0.21.1 The Differential

**Definition 0.21.8** (Differentiability by a Differential). *Let  $f$  be defined on a neighbourhood of  $c \in \mathbb{R}$ . The function  $f$  is differentiable at  $c$  in the differential sense if there exists a linear map  $L : \mathbb{R} \rightarrow \mathbb{R}$  such that*

$$f(c+h) - f(c) = L(h) + o(h) \quad \text{as } h \rightarrow 0.$$

*In that case the differential of  $f$  at  $c$  is  $df_c := L$ .*

**Theorem 0.21.9** (Differential and Derivative Agree). *Let  $f$  be defined on a neighbourhood of  $c \in \mathbb{R}$ . Then  $f$  is differentiable at  $c$  in the ordinary derivative sense if and only if it is differentiable at  $c$  in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

*Go to proof.*

**Theorem 0.21.10** (Uniqueness of the Differential). *If  $L_1, L_2 : \mathbb{R} \rightarrow \mathbb{R}$  are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \quad \text{as } h \rightarrow 0,$$

*then  $L_1 = L_2$ . Go to proof.*

**Theorem 0.21.11** (Differential Continuity Criterion). *If  $f$  is differentiable at  $c$  in the differential sense, then  $f$  is continuous at  $c$ . Go to proof.*

**Theorem 0.21.12** (Chain Rule for Differentials). *If  $f$  is differentiable at  $c$  and  $g$  is differentiable at  $f(c)$ , then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

*Go to proof.*

**Theorem 0.21.13** (Linearity of the Differential). *If  $f$  and  $g$  are differentiable at  $c$  and  $\alpha, \beta \in \mathbb{R}$ , then*

$$d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c.$$

*Go to proof.*

## Proofs